

CabinSentinel Wireframe & UI Flow

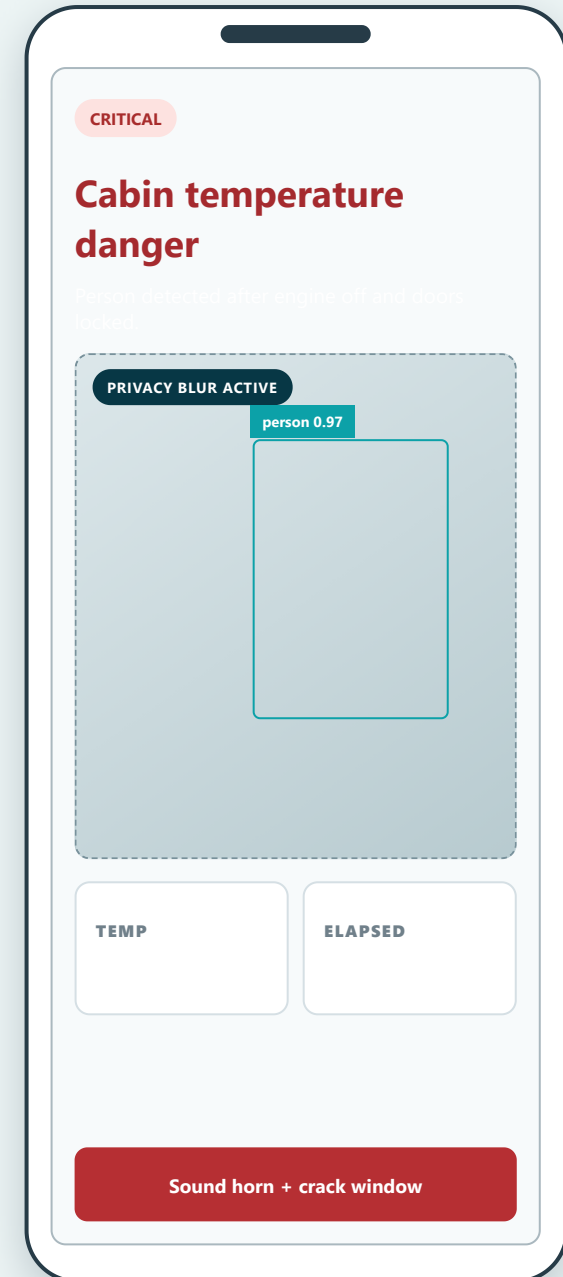
A safety-first owner console for understanding cabin risk, acting within seconds, and preserving privacy at the edge.

Team G05 · T068 · Gradflow

Artifact Detailed wireframe and interaction specification

Platforms Desktop dashboard + mobile alert

Version Gate G1 · 20 Sep 2026



1. Experience strategy and information architecture

PRINCIPLE 01

Safety before convenience

Life-critical state always wins visual priority, focus, audio and action placement.

PRINCIPLE 02

Explain every state

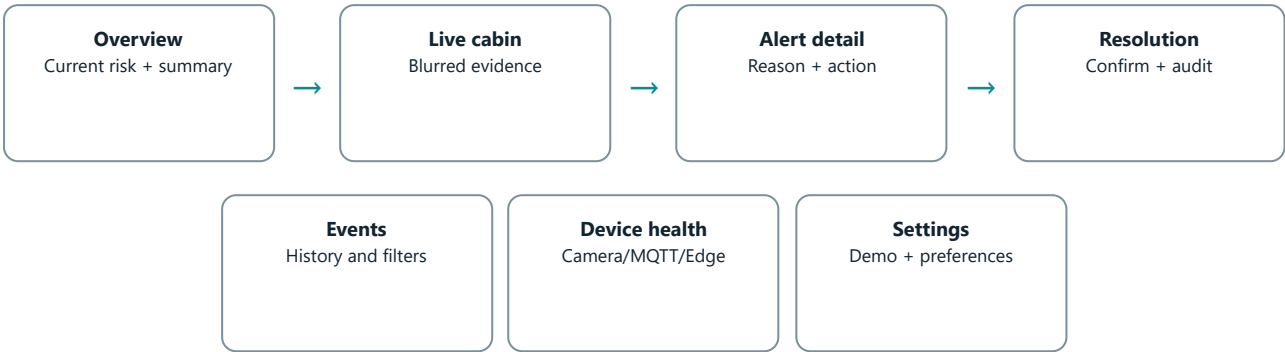
Show object, sensor evidence, rule and timestamp - never only a color.

PRINCIPLE 03

Private by default

Every visible frame is blurred and clearly labeled as processed locally.

Information architecture



State language

State	User meaning	Surface	Primary action
SAFE	No active risk	Persistent badge	View evidence
NOTICE	Possible forgotten item	Banner + push	Mark checked
WARNING	Occupant remains	Sticky alert	Check cabin
CRITICAL	Heat danger	Focus-lock modal	Emergency action
ERROR	Unable to verify	Health alert	Troubleshoot

A Priority order: device fault affecting safety → CRITICAL → WARNING → NOTICE → SAFE. The UI never infers risk; it renders the backend state and rule ID.

2. Visual system and reusable components

Color tokens



Brand / Action · #0D9CA3



Critical · #D94A4A



Warning · #E7A12E



Safe · #2AA66A



Error · #45545E

Typography and spacing

32 / 36

DISPLAY TITLE · SEMIBOLD

Section heading 20 / 24

Body copy 14 / 20. Minimum UI caption 12 / 16.

4 px base

8 px rhythm

12 px radius

44 px target

Critical copy uses direct verbs and measurable evidence. Avoid vague text such as "Something happened".

Core components

CRITICAL

Risk badge

Icon + text + color

Sensor tile

Label, current value, freshness

Action button

Idle / loading / success / failed

Evidence card

Blurred image + rule + timestamp

Event row

State transition + expandable detail

1 Focus

One dominant action per risk state.

2 Feedback

Every tool shows progress and result.

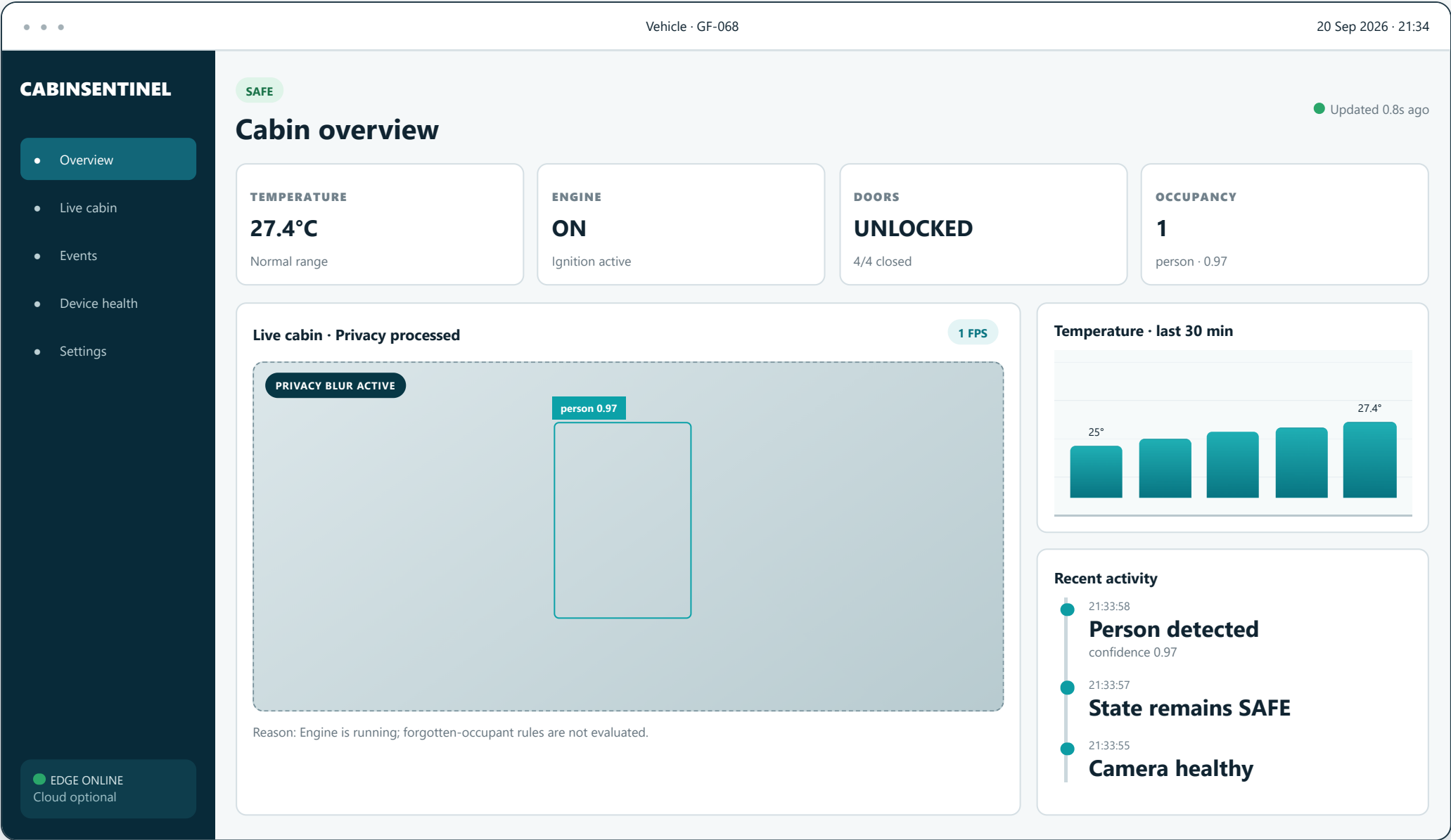
3 Recovery

Failure includes a safe next step.

4 Audit

All acknowledgements are logged.

3. Desktop overview - SAFE monitoring



1 State and freshness remain visible above the fold.

2 Evidence frame is explicitly labeled as blurred.

3 Reason sentence explains why the state is SAFE.

4 Cloud status never overrides Edge status.

4. WARNING - pre-critical intervention

SCREEN · 04

CABINSENTINEL

- Overview
- Live cabin
- Events
- Device health
- Settings

EDGE ONLINE

Internet disconnected · safety active

Vehicle · GF-068

20 Sep 2026 · 21:34

WARNING

Rule R-WARN-01 · 21:34:08

Occupant remains in parked vehicle

TEMPERATURE

32.0°C

Rising +1.3°/5m

ENGINE

OFF

42 sec ago

DOORS

LOCKED

All doors

OCCUPANCY

PERSON

3/3 frames

Evidence

Blurred locally

PRIVACY BLUR ACTIVE

person 0.97

Why this warning

Engine is off, doors are locked and a person remains inside. Temperature is below the 35°C critical threshold.

Check cabin now

Snooze 60 seconds

Next escalation

Automatically changes to CRITICAL if temperature reaches 35°C or risk increases.

1 Threshold and trend are shown together.

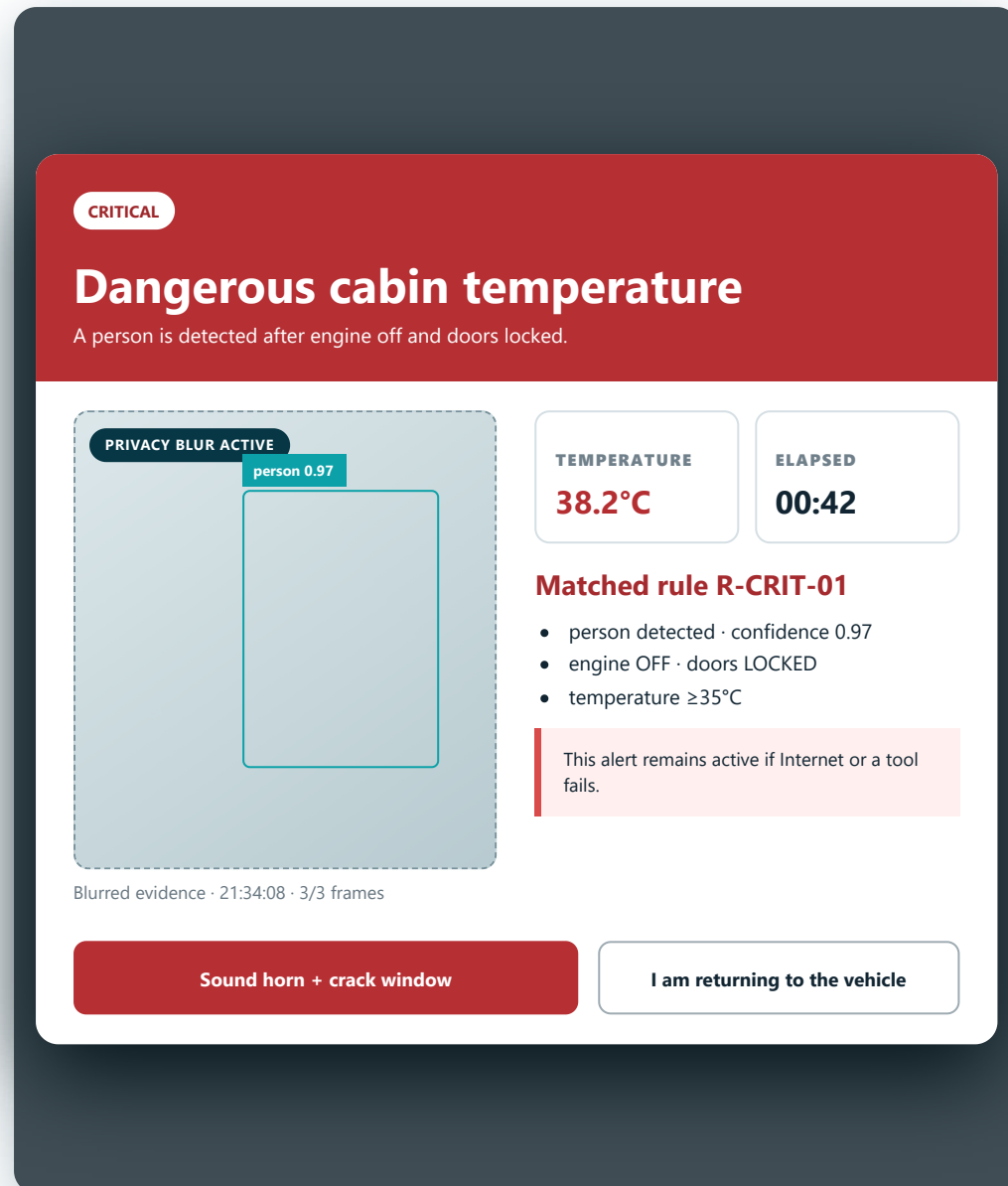
2 User sees the next escalation condition.

3 Snooze never suppresses continued evaluation.

4 Offline status reassures that safety still works.

5. CRITICAL - human-in-the-loop alert

SCREEN · 05



Interaction contract

1. Modal opens immediately and traps keyboard focus.
2. Screen reader announces state, reason and temperature.
3. Escape cannot dismiss a life-critical alert.
4. Primary CTA names the exact automated actions.
5. Repeated taps are disabled while tools run.
6. Acknowledgement requires a resolution reason.

Content hierarchy

- 1 Severity and plain-language danger
- 2 Evidence and matched rule
- 3 Temperature and elapsed time
- 4 Primary action and fallback

Fail-safe behavior

Partial failure keeps the alert CRITICAL and exposes a manual fallback. The UI never displays "resolved" solely because the camera feed disappears.

6. Action progress, partial failure and resolution

CABINSENTINEL

• Overview

• Live cabin

• Events

• Device health

• Settings

● EDGE ONLINE

Cloud unavailable · local queue

Vehicle · GF-068

20 Sep 2026 · 21:34

CRITICAL · ACTIVE

Event EVT-20260920-0042

Emergency actions in progress

Tool execution

✓

Vehicle horn triggered

Success · 21:34:09 · 184 ms

✓

Window cracked 4 cm

Success · 21:34:10 · 612 ms

...

Owner notification

Sending locally queued alert

!

SMS unavailable

No network · will retry

Confirm resolution

Choose only after physically checking the cabin.

Cabin is clear

Occupant is with me

Keep alert active

Closing the modal does not erase the event. Input, rule, state transitions, tool results and acknowledgement are retained locally.

Live risk timeline

Time	State	Reason	Action
21:33:26	WARNING	Person + engine OFF + 32°C	Owner notified
21:34:08	CRITICAL	Temperature reached 38.2°C	HITL modal
21:34:09	CRITICAL	Tool execution	Horn success

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7. Event history and explainable audit trail

CABINSENTINEL

• Overview

• Live cabin

• Events

• Device health

• Settings

EDGE ONLINE

Cloud optional

Vehicle · GF-068

20 Sep 2026 · 21:34

Event history

Auditable record of detections, decisions and user actions

All states ▾

Last 24 hours ▾

Export JSON

CRITICAL

1

WARNING

2

NOTICE

3

ERRORS

0

Time	State	Trigger	Evidence	Action result	Acknowledged by
21:34:08	CRITICAL	R-CRIT-01	person 0.97 · 38.2°C	Horn ✓ · Window ✓ · SMS queued	Owner · 21:35:02
21:33:26	WARNING	R-WARN-01	person 0.95 · 32.0°C	Push delivered	Escalated automatically
18:12:14	NOTICE	R-NOTICE-01	backpack 0.88	Reminder delivered	Owner · checked
08:44:39	SAFE	R-SAFE-01	engine ON	Log only	System

Expanded event · EVT-20260920-0042

Input snapshot:

camera OK, person 0.97, engine OFF, doors LOCKED, 38.2°C.

Decision:

WARNING → CRITICAL in 1.34 s.

Privacy:

frame reference points to blurred derivative only.

Audit requirements

• Immutable event ID and ISO timestamp.

• Previous/new state and rule version.

• Per-tool latency and response.

• User acknowledgement reason.

• No raw face frame in logs or export.

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8. Device health and error recovery

SCREEN · 08

CABINSENTINEL

- Overview
- Live cabin
- Events
- Device health
- Settings

EDGE DEGRADED

Camera blocked · sensors active

Vehicle · GF-068

20 Sep 2026 · 21:34

ATTENTION

Device health

Self-check completed 21:36:00

EDGE AGENT

Healthy

State machine · 22 ms

CABIN CAMERA

Blocked

No usable frame for 8 seconds

MQTT SENSORS

Healthy

Last event 0.4 s ago

Camera status · BLOCKED

PRIVACY BLUR ACTIVE

NO VERIFIABLE CABIN IMAGE

The system will not infer SAFE while occupancy cannot be verified.

Recovery checklist

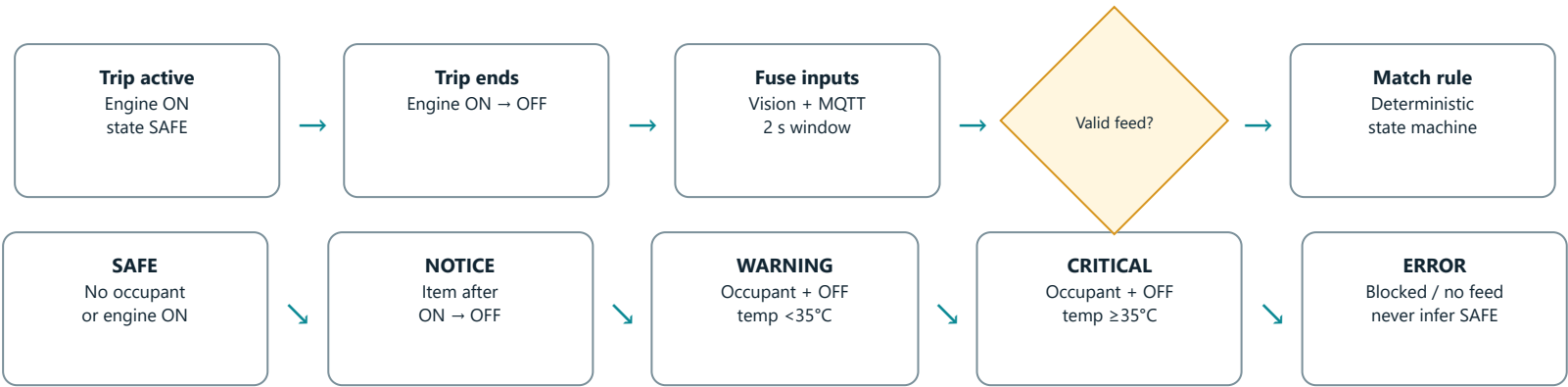
1. Check for objects covering the cabin camera.
2. Clean the lens and confirm the privacy cover is open.
3. Restart the camera module if the feed does not return.
4. Physically inspect the cabin before leaving.

Run camera self-test

Safe fallback

Sensor monitoring continues. Existing CRITICAL/WARNING events stay active until explicitly resolved.

9. End-to-end state and interaction flow



NOTICE branch

Push/banner → user marks checked or snoozes → event remains auditable.

WARNING branch

Sticky alert → continuous temperature evaluation → auto-escalate at threshold.

CRITICAL branch

Focus-lock modal → local tools → per-tool result → physical check → acknowledged resolution.

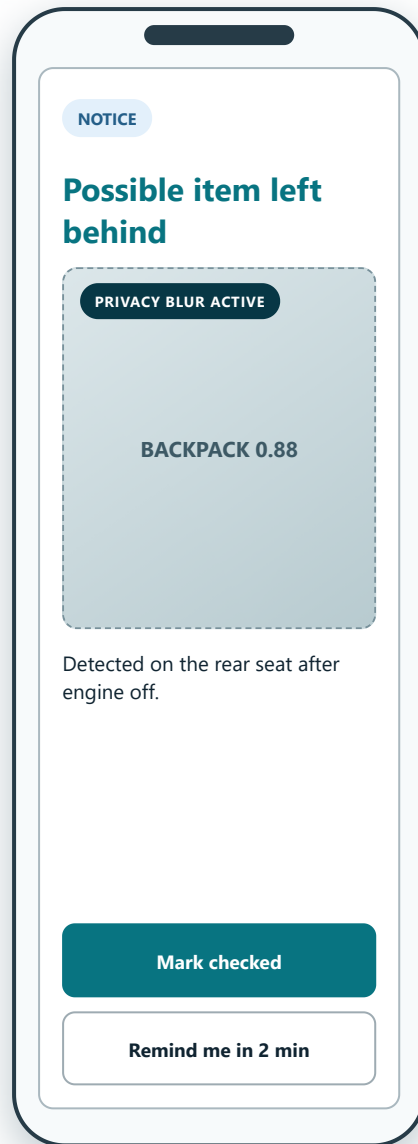
Critical sequence timing budget

Stage	Owner	Target	Visible UI	Failure behavior
Frame + sensor ingestion	Perception / MQTT	<700 ms	Data freshness	ERROR if stale
Rule evaluation	State machine	<300 ms	State transition	Keep previous risk
UI delivery	API / React	<250 ms	Modal + audio	Local retry
Tool execution	Action layer	Per-tool result	Progress rows	Expose fallback

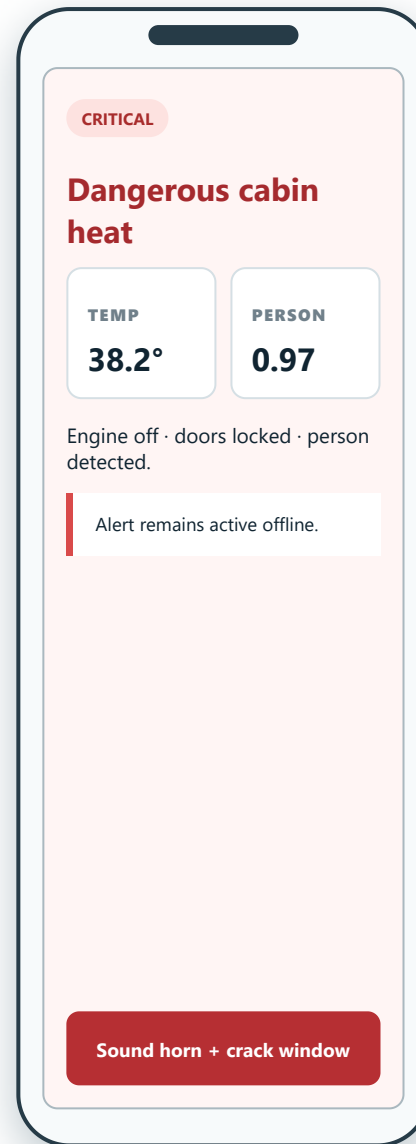
Non-negotiable: the LLM may rewrite alert copy, but it never assigns the safety state or blocks a deterministic action.

10. Mobile behavior, accessibility and developer handoff

HANDOFF · 10



NOTICE · non-blocking



CRITICAL · sticky action

Responsive rules

- Desktop 3-column layout collapses to state → evidence → actions → log.
- Critical action stays sticky above the mobile safe area.
- Temperature, timestamp and object never hide on small screens.

Accessibility acceptance

- Body contrast $\geq 4.5:1$; large text $\geq 3:1$.
- 44×44 px minimum target.
- State announced through ARIA live region.
- Modal focus trap; Escape disabled for CRITICAL.
- Color always paired with icon and label.

Developer handoff

1440 desktop

390 mobile

8 px grid

WCAG AA

UI receives typed state, rule ID, evidence metadata and tool status. It never calculates risk independently.